Kernix 0.11.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

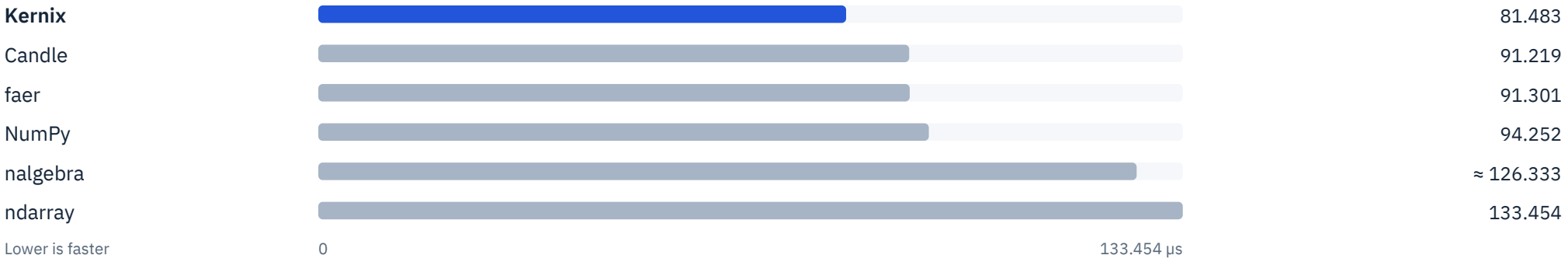
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

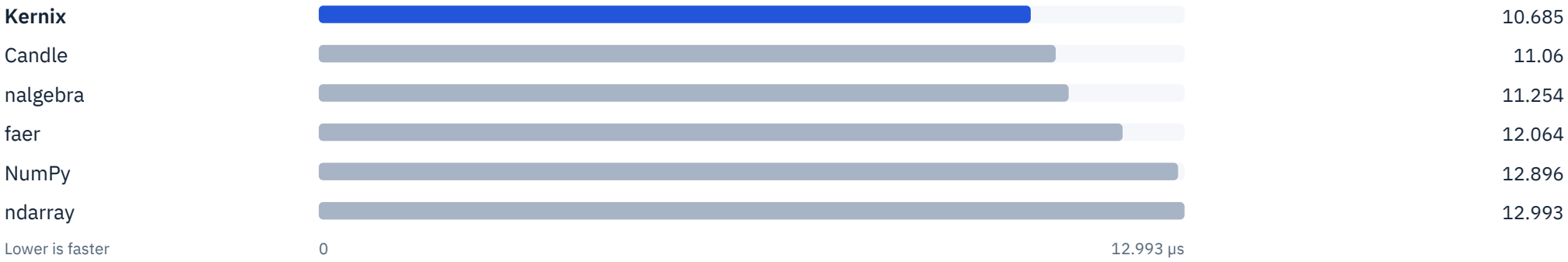
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



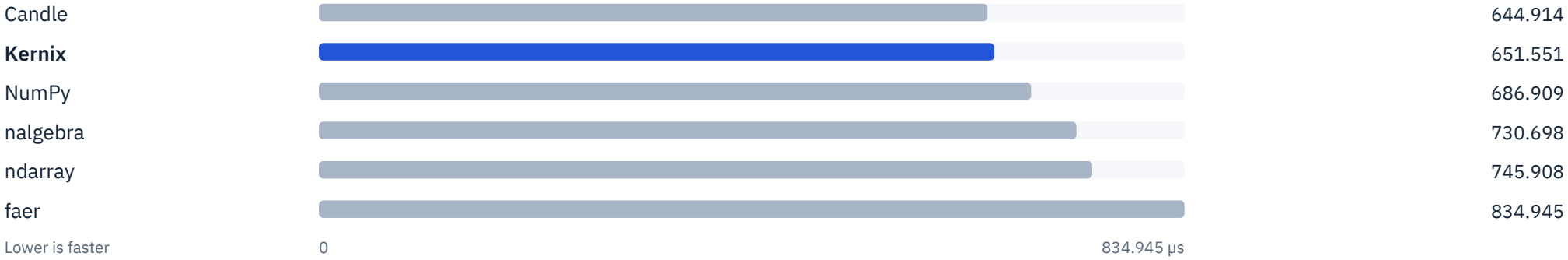
Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

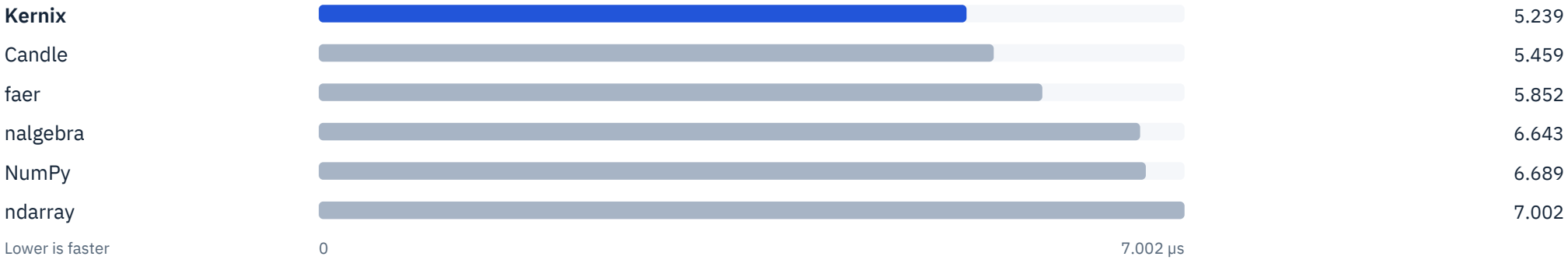
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



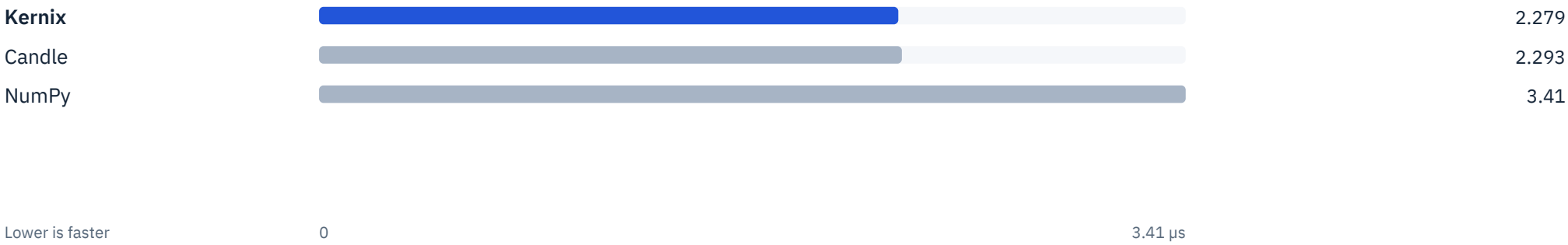
Kernix 0.11.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



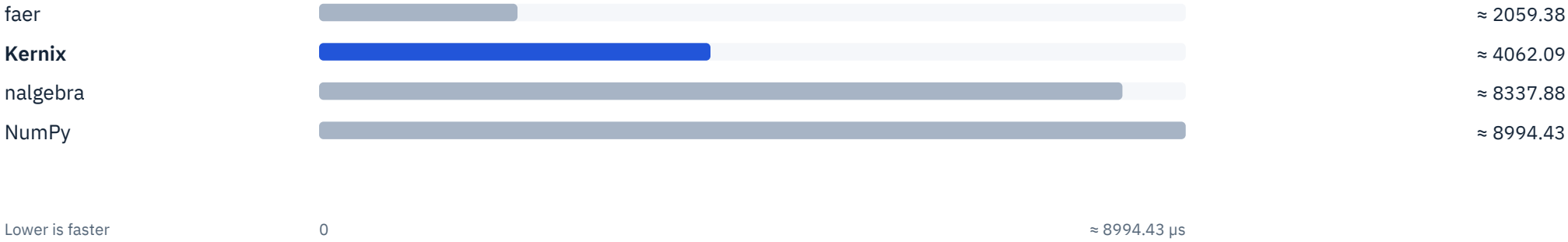
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



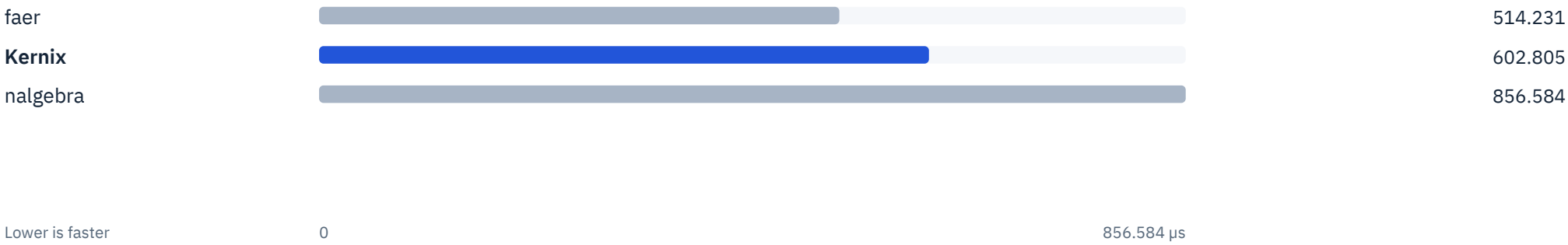
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

Per-call median

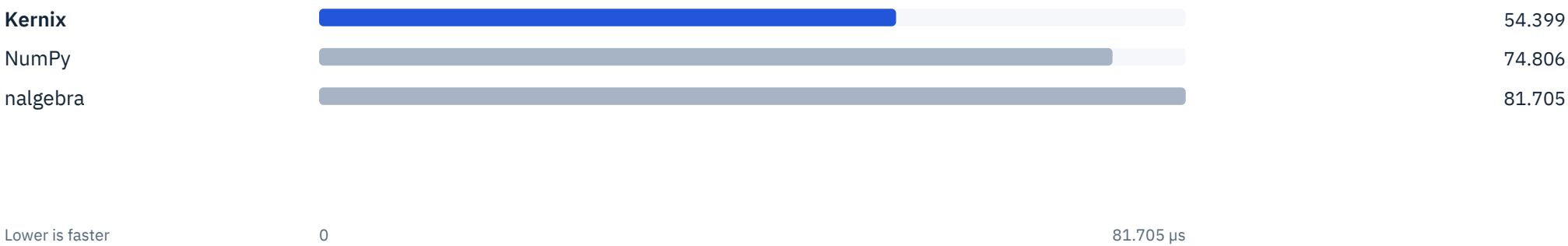
256 × 64 · f64 · 1 worker · allocated output



Minimum-norm least squares

Per-call median

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



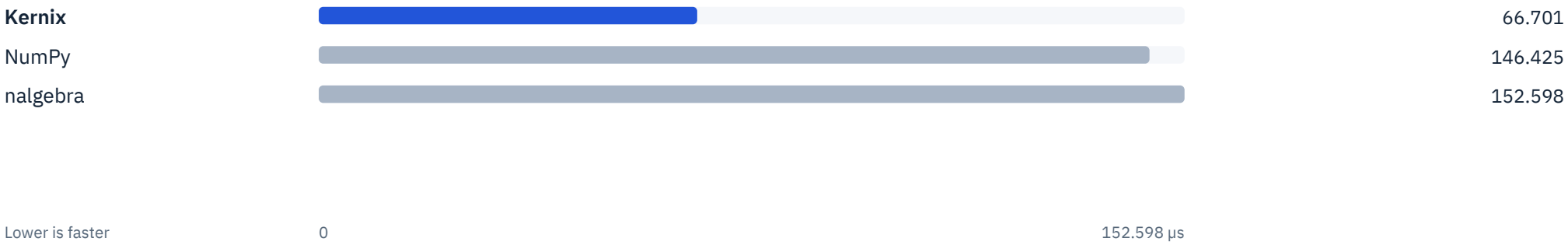
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

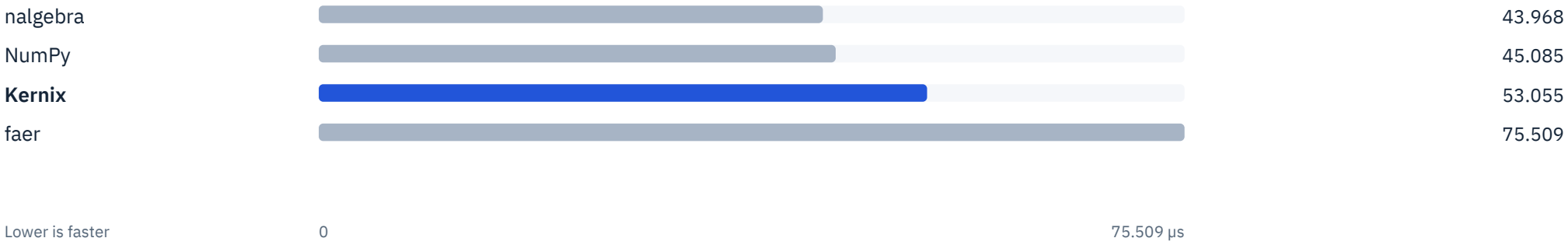
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

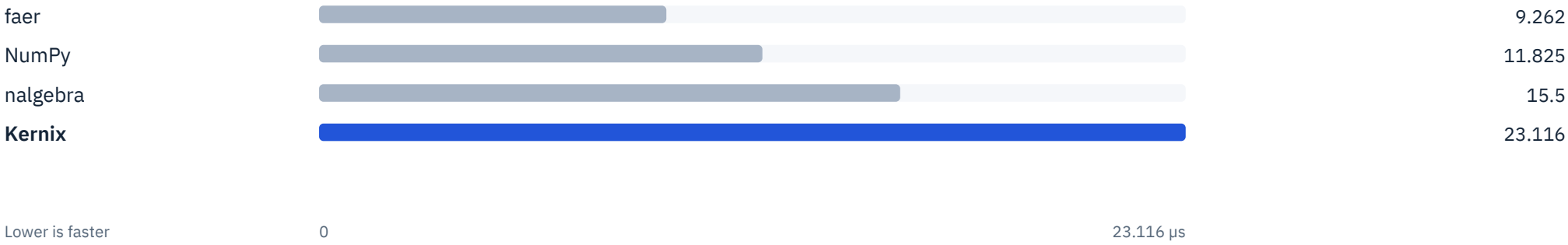
Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



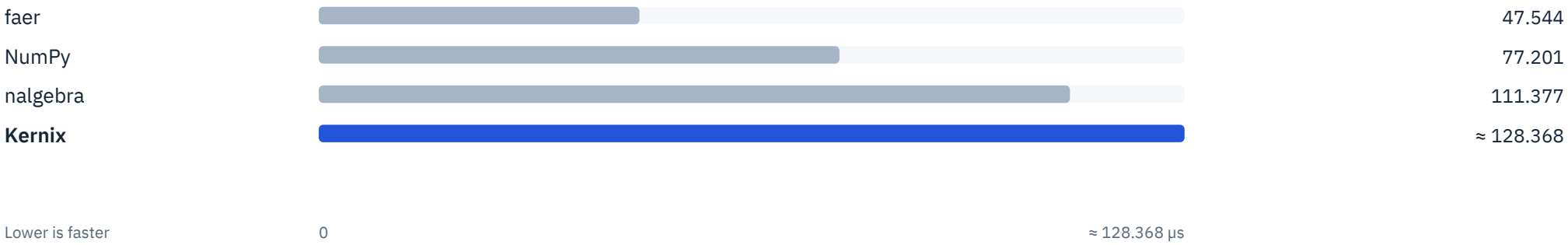
Kernix 0.11.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

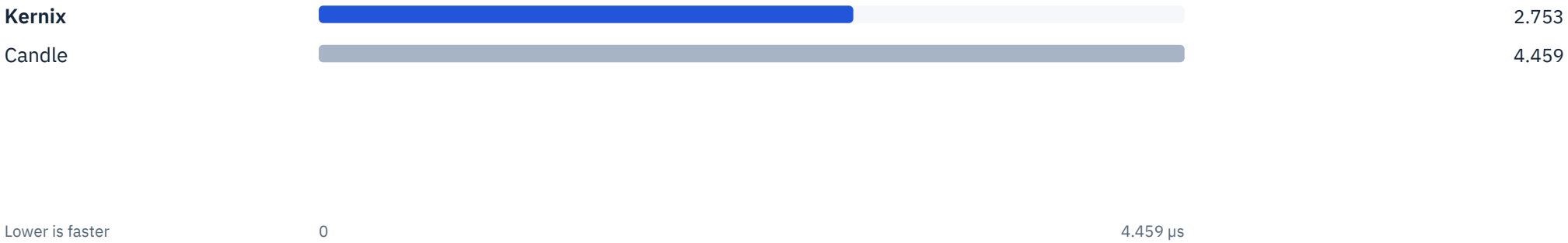
sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

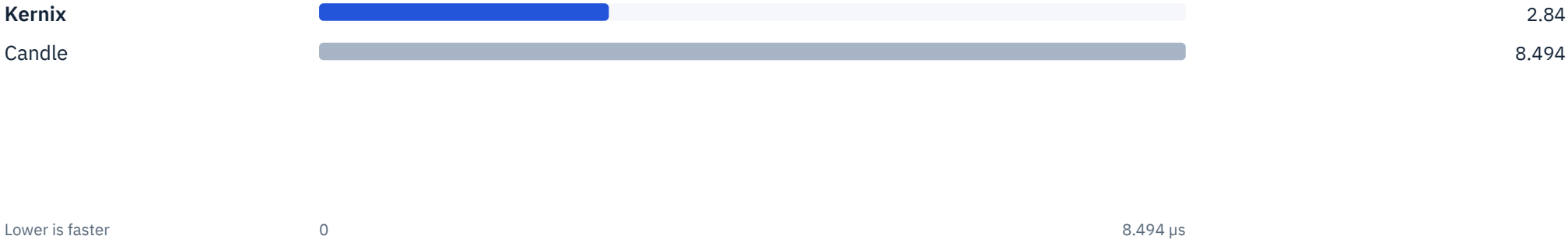
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



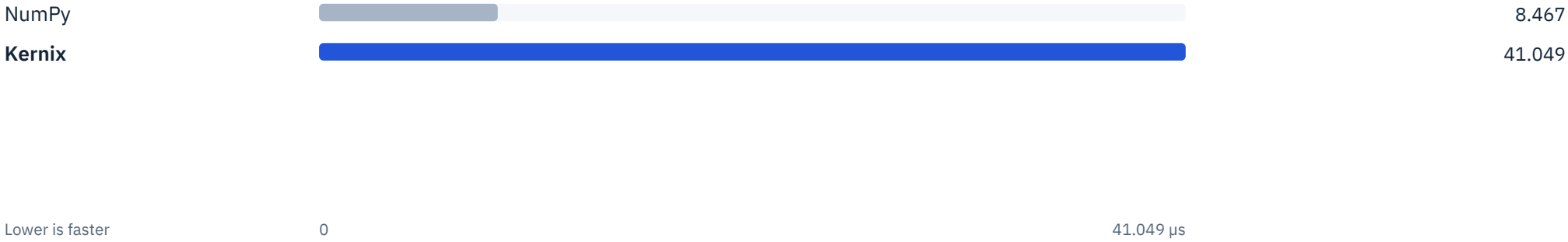
Kernix 0.11.0 / Tensor contractions

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median

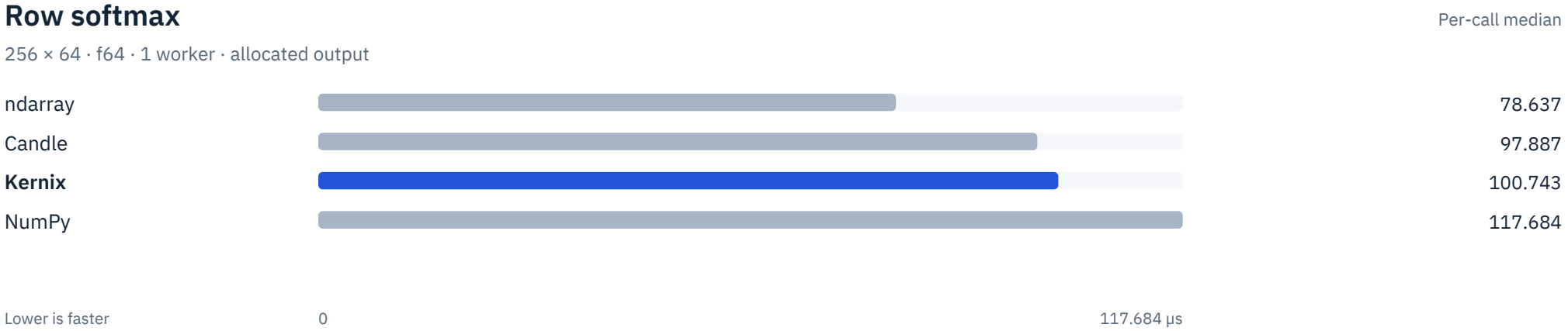


Kernix 0.11.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



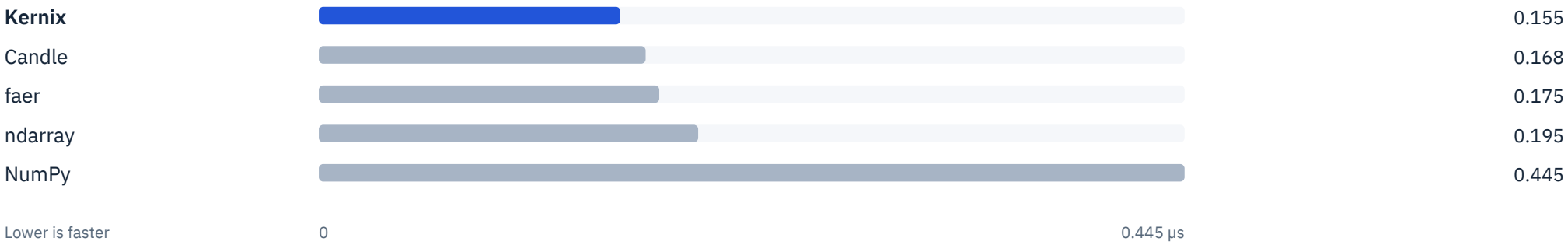
Kernix 0.11.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0025052882854137	0.9880735938161069 – 1.0225046982059274

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0289899805308893	0.9526051404500709 – 1.0830772982199464

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0114855606885347	0.990548270142216 – 1.0288933273574434

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0025730309864649	0.9680604337742499 – 1.0144496787739743

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.004740888037442	0.9773611561443327 – 1.030975013083748

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9811129303784333	0.963422583838335 – 1.0074494501596312

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0334319165802792	1.0075385294162393 – 1.0426613786824277

Cholesky factorization

256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7762255134210398	0.7593444192198485 – 0.8109318717436064

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.5982396425193351	0.5947366010193733 – 0.6118751691686888

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.872338778354107	0.5543263964950712 – 1.3800706430704246

Version comparison / paired processes

12th Gen Intel(R) Core(TM) i3-1215U · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0384417006440088	0.9568996441051641 – 1.3675399257855017

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9749494851133238	0.5880299992574441 – 1.1664146010099716

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	79522	79.522
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	82316	82.316
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	41398	41.398
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	40851	40.851
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1752	1.752
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4777	4.777
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1610	1.61
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1693	1.693
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1646	1.646
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1822	1.822

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	12993	12.993
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	11060	11.06
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	10685	10.685
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11254	11.254
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12896	12.896
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	12064	12.064
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	94252	94.252
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	91219	91.219
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	126333.00000000001	126.33300000000001
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	91301	91.301

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	133454	133.454
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	81483	81.483
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	651551	651.551
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	686909	686.909
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	730698	730.698
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	745908	745.908
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	644914	644.914
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	834945	834.945
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6643	6.643
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	5459	5.459

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5852	5.852
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7002	7.002
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5239	5.239
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6689	6.689
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	46010	46.01
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	54284	54.284
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	41717	41.717
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	41294	41.294
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	54141	54.141
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	47184	47.184

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	346674	346.674
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	333271	333.271
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	453024	453.024
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	311220	311.22
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	358189	358.189
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	446141	446.141
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2293	2.293
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3410	3.41
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2279	2.279
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	11276	11.276

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	79016	79.016
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	343076	343.076
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	41048	41.048
Cholesky factorization / nalgebra 512 × 512 · f64	8337880.000000001	8337.880000000001
Cholesky factorization / NumPy 512 × 512 · f64	8994431	8994.431
Cholesky factorization / faer 512 × 512 · f64	2059378.0000000002	2059.3780000000002
Cholesky factorization / Kernix 512 × 512 · f64	4062093	4062.093
Column-pivoted QR / Kernix 96 × 32 · f64	61850	61.85
Column-pivoted QR / faer 96 × 32 · f64	81252	81.252
Column-pivoted QR / nalgebra 96 × 32 · f64	84895	84.895

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	514231	514.231
Column-pivoted QR / nalgebra 256 × 64 · f64	856584	856.584
Column-pivoted QR / Kernix 256 × 64 · f64	602805	602.805
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	81705	81.705
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	74806	74.806
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	54399	54.399
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	152598	152.598
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	146425	146.425
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	66701	66.701
Thin QR / Kernix 96 × 32 · f64	53055	53.055

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	75509	75.509
Thin QR / NumPy 96 × 32 · f64	45085	45.085
Thin QR / nalgebra 96 × 32 · f64	43968	43.968
Thin QR / faer 256 × 64 · f64	508628	508.628
Thin QR / Kernix 256 × 64 · f64	588471	588.471
Thin QR / NumPy 256 × 64 · f64	430596	430.596
Thin QR / nalgebra 256 × 64 · f64	434514	434.514
Cholesky factorization / NumPy 64 × 64 · f64	11825	11.825
Cholesky factorization / nalgebra 64 × 64 · f64	15500	15.5
Cholesky factorization / faer 64 × 64 · f64	9262	9.262

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	23116	23.116
Cholesky factorization / Kernix 128 × 128 · f64	128367.99999999999	128.36799999999999
Cholesky factorization / nalgebra 128 × 128 · f64	111377	111.377
Cholesky factorization / faer 128 × 128 · f64	47544	47.544
Cholesky factorization / NumPy 128 × 128 · f64	77201	77.201
Cholesky factorization / faer 256 × 256 · f64	227006	227.006
Cholesky factorization / Kernix 256 × 256 · f64	579500	579.5
Cholesky factorization / nalgebra 256 × 256 · f64	774853	774.853
Cholesky factorization / NumPy 256 × 256 · f64	605474	605.474
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4717	4.717

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6946	6.946
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2753	2.753
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4459	4.459
Model inference / Candle batched normalized/sliced MLP · f64	4733	4.733
Model inference / Kernix batched normalized/sliced MLP · f64	4036	4.036
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2840	2.84
Model recording + forward / Candle batched normalized/sliced MLP · f64	8494	8.494
Model backward / Kernix prepared graph → fresh gradients · f64	3515	3.515
Model backward / Candle prepared graph → fresh gradients · f64	21894	21.894
Parameter update / Kernix transactional SGD, fixed gradients · f64	584	0.584

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	41049	41.049
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	8467	8.467
Row softmax / Candle 256 × 64 · f64	97887	97.887
Row softmax / ndarray 256 × 64 · f64	78637	78.637
Row softmax / Kernix 256 × 64 · f64	100743	100.743
Row softmax / NumPy 256 × 64 · f64	117684	117.684
Create zeros / Kernix 64 × 64 · f32	155	0.155
Create zeros / faer 64 × 64 · f32	175	0.175
Create zeros / ndarray 64 × 64 · f32	195	0.195
Create zeros / Candle 64 × 64 · f32	168	0.168

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	445	0.445
Create ones / Candle 64 × 64 · f32	250	0.25
Create ones / NumPy 64 × 64 · f32	1399	1.399
Create ones / Kernix 64 × 64 · f32	205	0.205
Create ones / ndarray 64 × 64 · f32	169	0.169
Create ones / faer 64 × 64 · f32	207	0.207